19. Application of fn and Inverse function

① Normalization and Standardization

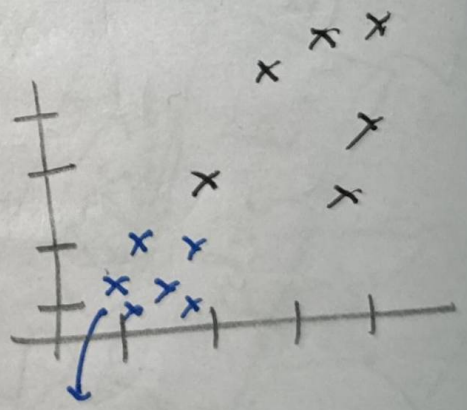
(Independent features) | (Dependent features)

Carpet Area (Sq.ft)	No. of Rooms	Area (Sq.ft)	Price (INR/USD)
1500	4	1800	50 lakhs
1800	3	2100	60 lakhs

↓

if we scale down all these values to a smaller scale

if we have huge magnitudinal values, it will take time obviously



After norm, we shrink them b/w (0 & 1)

CA →
NR →
Area →

$\boxed{\text{ML Model}}$ → o/p ⇓ price

{mathematical operation} ⇒ happen quickly

In Normalization

initially $\boxed{1500 \times 1500}$

↓

is reduced to $\boxed{1.5 \times 1.5}$ → for easy operation

Standardization (Scaling down)

Not done on Dependant feature

We take the Data feature → Transform them in a way that $\boxed{\mu = 0 \ \& \ \sigma = 1}$ called as Standard Normal Distribution (good for optimisation process)

μ → Mean, σ → S.D

No. of Rooms

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \xrightarrow{f(x)} \begin{bmatrix} -1.5/2 \\ -0.5/2 \\ 0.5/2 \\ 1.5/2 \end{bmatrix}$$

$f^{-1}(x)$ Inverse fn.

No. of Rooms'

$$Z_1 = \frac{1-\mu}{\sigma}$$

Eg

① $\frac{1-2.5}{2} = \frac{-1.5}{2}$

(Similarly for others)

$$\mu = \frac{1+2+3+4}{4} = \frac{10}{4} \rightarrow \boxed{2.5}$$

$\boxed{\sigma = 2}$ Assumption

original transformation $(z) = \frac{x-\mu}{\sigma}$

Inverse transformation $(x) = z\sigma + \mu$

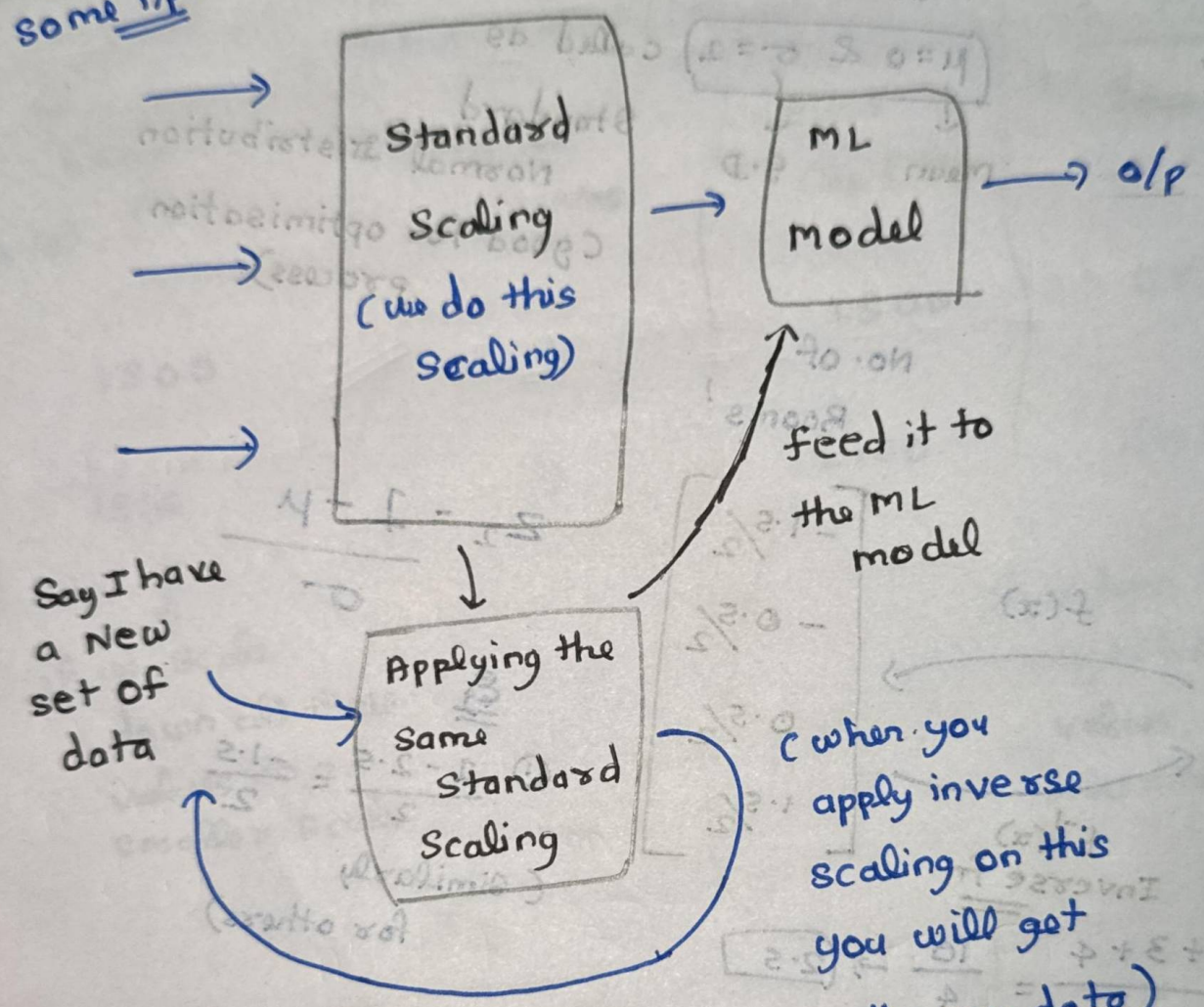
Applying Inverse

$$x = \left(\frac{-1.5}{2}\right) \times 2 + (2.5)$$

$$x = -1.5 + 2.5$$

$$\boxed{x = 1}$$

what is the use of Inverse transformation



usecase

(in the above img)

along with i/p

depend o/p

(if we scall the o/p too) → then our prediction will also be scaled (Scaled o/p) in order to get the original o/p here you have to ~~scale to~~ do inverse scaling)

f_1 f_2 f_3 o/p

⇓

Standard Scaling

Price o/p → [Model Training (we train the model)] → o/p [Scaled o/p]

To find the original one, we need to do the $\boxed{\text{Inverse}}$

if so we feed new i/p to it → [Again we should scale them]

Normalization

Feature Scaling with Min Max Normalisation

→ original Transformation

$$z = \frac{x - \min(x)}{\max(x) - \min(x)} \rightarrow \boxed{T: x \rightarrow y}$$

Inverse Transformation

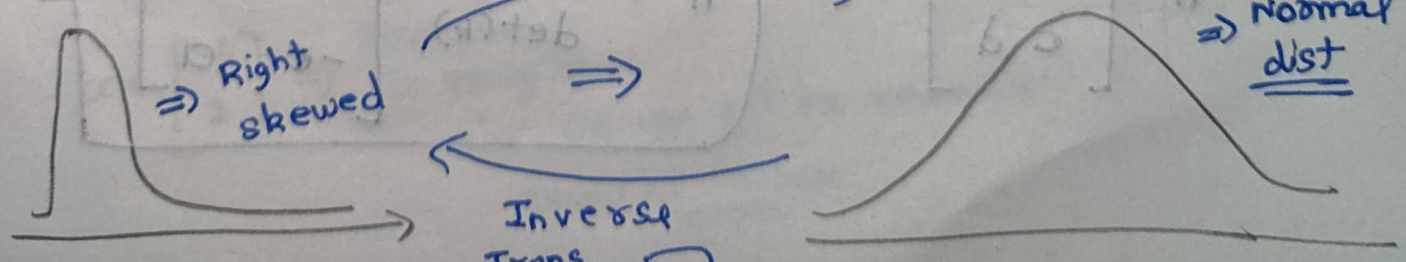
$$x = \frac{z\,(\max(x) - \min(x))}{x - \min(x)}$$

↳ $\boxed{T^{-1}: y \rightarrow x}$

Distribution of Data

original

Logarithmic Distribution Trans → $\boxed{y = \log(x)}$ Normal Distribution



Inverse Trans $\boxed{x = e^y}$

(5) Data Encryption and Decryption

Enfn:- $E(p) = c$ $\quad p \rightarrow$ plain text

Dfn $\rightarrow D(c) = p$ $\quad c \rightarrow$ cipher text

Use case

Sensitive data

E $\rightarrow$ when storing

D $\rightarrow$ when viewing back info